

Ajay J

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Summary

AI/ML engineer with 12+ months production experience in retrieval-augmented generation (RAG), multi-agent systems, and semantic search architectures. Specialized in deploying cost-optimized generative AI solutions on resource-constrained infrastructure. Shipped 5+ production AI systems with 95-96% accuracy metrics.

Experience

AI Engineer Intern (Volunteering), *Karma Com Solidarité (KCS)* Dec 2025 – May 2026

- Built role-based RAG assistant for volunteer tracking (50+ users) with N8N, Qdrant, LangGraph; 500ms latency on 10K+ documents
- Architected secure multi-tier access controls (Admin/Employee/Volunteer) with on-premise GitLab/Wiki integration
- Tools:** Ollama, N8N, Qdrant, LangGraph, Docker

Product Development Intern, *Stellantis N.V* June 2025 – Dec 2025

- Engineered multi-format document parameter extraction engine (PDF/DOCX/XLSX/XML) with SpaCy NLP; 96% accuracy validated by QA
- Built Streamlit UI with dark mode, filtering, CSV export for business users; Won 1st Place in VPE Hackathon
- Tools:** Ollama, Qwen 1.5B-Thinking, Streamlit

AI/ML Software Engineer Intern, *Wikasta Technical and Business Solutions* June 2025 – July 2025

- Project 1:** Duplicate Detection Pipeline — multi-stage verification (SQL + LLM + vector search) achieving 95% accuracy on 100-question production dataset
- Project 2:** Resume Parser — OCR + unified LLM stack reducing inference overhead by 40% vs dual-model approach
- Tools:** Ollama, PostgreSQL + PgVector, Qwen-2.5 Thinking, Docker

AI Engineer Intern, *Zetheta Algorithms* June 2025 – July 2025

- Architected multi-agent LangGraph evaluation framework automating LLM assessment across accuracy, clarity, pedagogy, tone metrics using Evidently library
- Engineered real-time fact verification (DuckDuckGo + LLM evidence analysis); deployed for persona-driven chatbot development
- Tools:** LangGraph, Smolagents, Google ADK, DuckDuckGo Search API

AI/ML Engineer Intern, *DYNX Technologies* July 2024 – Sept 2024

- Designed virtual assistant for customer interactions with NLP-based intent recognition and entity extraction
- Tools:** NLTK, Chatterbot, Spacy

Data Scientist Intern, *Getmax Healthcare* Jan 2024 – May 2024

- Led team of 4 on healthcare analytics; integrated report generation and UI creation with PDF export
- Tools:** Dash, NumPy, Pandas, Scikit-learn

Projects

KCS Intelligence Hub: RAG Agent with Role-Based Access Jan 2026 – Feb 2026

- Built RAG assistant for Karma Com Solidarité enabling volunteer activity tracking; 50+ users, 500ms latency on 10K+ documents
- Deployed on-premise with GitLab and Wiki integration using N8N workflow automation and Qdrant vector DB
- Tech:** Qdrant, Docker, N8N, LangGraph

PersonaPulse: Multi-Agent Chatbot Evaluation Framework June 2025 – July 2025

- Multi-agent LangGraph workflow automating evaluation across accuracy, clarity, pedagogy, tone; Evidently library integration

- Real-time fact verification combining DuckDuckGo search with LLM evidence analysis; deployed for Zetheta product team
- **Tech:** LangGraph, Qdrant, DuckDuckGo API, Evidently

Subject-Expert Chatbot for Physics/Research

Jan 2025 – Feb 2025

- RAG chatbot trained on 2500+ solved physics(fluid mechanics) problems with Chroma DB and Gemini embeddings; 50+ user sessions, 87% relevance scoring
- Multi-modal interface: text + image upload (OCR + vision model) for diagram-based problem solving; Gradio UI deployed on Colab
- Refactoring to LangGraph agent (removing LangChain) for autonomous web search and 35% token overhead reduction
- **Tech:** LangChain, Chroma, HuggingFace Transformers, Gradio, Llama-3.2-11B-Vision

Skills

Languages: Python (Advanced), Java (Intermediate)

LLM & Orchestration: LangChain, LangGraph, Ollama, Smolagents, LlamaCpp

NLP & Semantic Search: NLTK, Spacy, Gensim, HuggingFace Transformers, TensorFlow, PyTorch

Vector Databases: Qdrant, Chroma, PostgreSQL + PgVector, DuckDuckGo Search API

Deployment: N8N, Docker, Streamlit, Gradio, FastAPI

Education

B.E. in Mechanical Engineering, St. Joseph's Institute of Technology
CGPA: 7.79/10

Nov 2022 – May 2026

Certifications & Achievements

NVIDIA: Building RAG Agents with LLMs

Nov 2025

ORACLE: Cloud Infrastructure Generative AI Professional 2025

Nov 2024

Top Performer: Stellantis VPE Hackathon (1st Place + Honorary Finalist)

June 2025 – Dec 2025

Top Problem Solver: AI Technology, Cogtis Technologies

Mar 2024 – Aug 2024